

Hieu Nguyen

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Summary

AI engineer with 4 years of experience building production systems across retrieval, NLP, speech recognition and computer vision. Hands-on CTO work at Plenarius and Sleep Sage spans Go and Python services, mobile interfaces and tested AI workflows. Previously worked on production AI at Viettel Post and FPT Software; currently researching tool-using agents at QUT. Completing a Master of Artificial Intelligence, with expected completion in November 2026.

Skills

- **AI/ML:** RAG, vector search and quantisation, agentic workflows, LLM orchestration, transformers, NLP, speech recognition, computer vision, anomaly detection.
- **Engineering:** Python, Go, TypeScript, PostgreSQL, pgvector, MongoDB, REST APIs, React, React Native, Expo, FastAPI, Docker, CI/CD, data pipelines.
- **Cloud & infrastructure:** Azure (Functions, Cosmos DB), AWS (EC2, S3), Hetzner, Cloudflare (WAF, R2 object storage), Caddy, systemd.

Experience

Chief Technology Officer

Sep 2026 – Present

Sleep Sage

- Lead technical development of an education app that turns PDFs and text into source-grounded concepts, user-approved learning cues and private speech clips.
- Improved React Native and Expo session editing, audio playback and first-night setup, and strengthened FastAPI processing with regression tests for validation, authentication and playback.

Chief Technology Officer

Apr 2026 – Present

Plenarius (formerly Calimac)

- Inherited a PHP-based platform and led its migration to Go and PostgreSQL, moving 96 server-rendered pages, 205 API paths and the recommendation system onto the new production stack with no PHP in the request path.
- Designed vector search over roughly 200,000 papers and evaluated retrieval configurations, balancing response quality, query cost and infrastructure trade-offs.
- Run the production platform on a single 8-vCPU server for roughly US\$43 per month all in, tracking retrieval quality, cost per query and infrastructure trade-offs.

AI Researcher

Jan 2026 – Jun 2026

Cognilaw

- Evaluated and improved a hallucination-resistant RAG pipeline for legal workflows, testing retrieval strategies, grounded generation and answer validation using retrieval and response-quality metrics.

Research Assistant

Sep 2025 – Present

Queensland University of Technology

- Evaluated and engineered an agentic person re-identification system combining appearance, body composition, gait and physical attributes through confidence-weighted voting.
- Cleaned and prepared biomedical literature datasets, then fine-tuned transformer models for protein-protein interaction relation extraction.

AI Software Engineer

Apr 2023 – Nov 2024

Viettel Post

- Integrated Whisper fine-tuned on Vietnamese with BERT classifiers for offensive-language detection and mood prediction, achieving 90% accuracy in production.
- Trained an Extra Trees classifier for customer churn prediction, reaching 80% accuracy.

AI Software Engineer

Oct 2021 – Mar 2023

FPT Software

- Contributed to FPT's [US-patented](#) Sound AI acoustic anomaly-detection platform, deployed on a Samsung factory floor. Built its Python backend on Azure Functions, supporting model serving, CI/CD and production monitoring.

Research Intern, Computer Vision

Apr 2021 – Jul 2021

Asilla

- Fine-tuned and deployed real-time [human pose estimation](#) on Jetson Nano with TensorRT, using INT8 quantisation and structured pruning to improve efficiency by 20% under edge-compute constraints.

Projects & Awards

Planner and executor collaboration

Manuscript in preparation

Stack: Python, AppWorld, BFCL, Agent evaluation

Co-author with Kien Nguyen Thanh at QUT of *How Should a Stronger Planner Help a Smaller Agent? Preregistered Tests of When to Hand Over and What to Send*. Built tool-using agent orchestration and paired replay evaluations comparing handover, advice and direct-action strategies with statistical analysis and cost-quality trade-offs.

Local AI Workflow Platform

Presented | AgentCamp 2026, Brisbane Edition

Stack: Python, LangGraph, FastAPI, React Flow, Pydantic, YAML

Local-first platform for authoring, running and visualising AI workflows from declarative YAML. Each workflow is validated through a Pydantic GraphSpec contract before being compiled into runnable graphs, with approval interrupts, rollback restore and eval coverage.

Capture the Narrative

1st Place | UNSW National AI Competition

Stack: LLMs, multi-agent simulation, AWS EC2, S3

Built generative-AI agents inside a simulated election environment as an adversarial exercise in how synthetic content shifts sentiment.

SafeMind

Most Viable | TANDA GenAI Hackathon

Stack: Python, Flask, OpenAI API, MongoDB, python-docx

Voice-driven workplace incident capture that pre-fills WorkCover claim forms, compressing hours of paperwork for injured workers.

Education

Master of Artificial Intelligence

2025 – 2026

Queensland University of Technology | GPA 6.538/7.0

Final semester, Semester 2 2026; expected completion November 2026.

Bachelor of Information Technology

2018 – 2021

University of Science and Technology of Hanoi

Volunteer

Vice President

Oct 2025 – Oct 2026

QUT The Emerging Coders Hub (TECH)

Ran coding and IT workshops teaching beginners to build chatbots, personal websites and AI agents, and organised career events including mock interview sessions, progressing from General Executive (Jun–Oct 2025).

Project Manager

Sep 2025 – Present

QUT AI & ML Society

Led 30+ members building a predictive analytics dashboard on real insurance data for claim-delay prediction and anomaly detection, stepping up from Project Officer (Aug–Sep 2025).